Use Case 1: Book Appointment

Scope: Appointment Tracking System

Primary Actor: Patient or Administrative Staff

Stakeholders and Interests:

- **Patient:** Want to make an appointment with the appropriate healthcare professional based on their availability
- **Healthcare Professional**: Wants to receive patients for treatment while making sure the appointments follow their schedule and availability
- **Adminitrative Staff:** Act as an intermediary between the patient and system to help the patient make an appointment with the appropriate healthcare professional based on the healthcare professional's availability.

Preconditions:

- The patient must be already registered into the system
- The patient or the administrative staff must be logged into the system and be authorized to make appointments
- The healthcare professional is already registered into the system
- The healthcare professional must have a schedule in the system
- The clinic location is already recorded into the system
- The medical service requested must already be defined in the system

Success Guarantee (Postconditions):

- The appointment is successfully created and recorded in the system.
- The appointment is associated with the patient, healthcare professional, requested medical service and clinic location
- The appointment is scheduled on a date and time that follows the healthcare professional's availability and applicable scheduling constraints.
- The patient has the referral if the medical service requires one
- The appointment is saved with the appropriate booking status.
- Authorized users can access the appointment information based on their roles.

Main Success Scenario (Basic Flow)

1. The patient or administrative staff member requests the option to book an appointment.
2. The system requests the information needed to book an appointment.
3. The system identifies the patient. If administrative staff is making the booking, they identify the patient, and the system retrieves the patient's information.
4. The user specifies the requested medical service.
5. The system provides the healthcare professionals and clinic locations associated with the requested service.
6. The user selects a healthcare professional and clinic location.
7. The system checks the professional's schedule, availability, scheduling constraints, and referral requirements.
8. The system provides the available appointment dates and times.

9. The user selects a date and time for the appointment.
10. The system verifies that the selected appointment satisfies the booking requirements and provides the appointment details.
11. The user submits the appointment booking.
12. The system creates and records the appointment with the patient, healthcare professional, medical service, clinic location, scheduled date and time, and appropriate status.
13. The system provides the successful booking result and appointment details.

Extensions (Alternative and Failure Flows)

7a) The requested medical service requires a referral, but the patient does not have the required referral.

1. The system informs the user that a medical referral is required to book the selected professionals for an appointment.
2. The system does not complete the appointment booking.
3. The use case ends without creating an appointment.

7b) The healthcare professional has no available appointment times.

1. The system informs the user that no appointment times are available for the selected healthcare professional.
2. The system allows the user to select a different healthcare professional or clinic location.
3. The user selects a different healthcare professional or clinic location.
4. The use case resumes at step 7 of the Main Success Scenario.

Special Requirements

- Only authorized patients and administrative staff can book appointments and access the appointment information they are permitted to view.
- The system must record relevant booking activities to support accountability and protect patient confidentiality.

Technology and Data Variations List

- The appointment can be booked directly by the patient or by administrative staff on behalf of the patient.
- The administrative staff can search the patient using their insurance number, date of birth, phone number, or partial first and/or last name.

Frequency of Occurrence

- Whenever a patient or administrative staff member needs to book an appointment

Miscellaneous

- **Open Issues:**
    - For uninsured services, does the patient need to pay when booking the appointment or after the consultation?

    - What should happen if the appointment booking fails? Should the system allow the patient to try another date or healthcare professional?
    - Does the system automatically confirm the appointment after booking, or does it require an additional confirmation step?
- **Assumptions:**
    - The patient is already registered in the system before booking an appointment.
    - The patient or administrative staff is logged into the system before booking an appointment.
    - The healthcare professional is already registered in the system and has an existing schedule.
    - The clinic location and medical services are already recorded in the system.
    - Patients can book appointments directly through the system, while administrative staff can book appointments on behalf of patients.
    - When an appointment cannot be booked, the system informs the user of the reason.
    - After booking, the system displays the appointment details to the user.

Use Case 2: Register Patient Arrival

Scope: Appointment Tracking System

Primary Actor: Administrative staff

Stakeholders and Interests:

- **Administrative Staff:** Wants to update the system to record that the patient has arrived for their appointment.
- **Patient:** Wants to confirm their arrival at the clinic for their scheduled appointment.
- **Healthcare Professional:** Wants to know when their patient has arrived and is ready for the consultation.

Preconditions:

- The patient must already be registered in the system.
- The patient must already have an appointment recorded in the system.
- The administrative staff member must be logged into the system and authorized to register patient arrivals.
- The patient must have arrived at the clinic for their appointment.

Success Guarantee (Postconditions):

- The patient's appointment is updated in the system to show that the patient has arrived.
- The updated arrival information is saved in the system.
- The arrival is recorded for the correct patient and their scheduled appointment.
- The healthcare professional and authorized administrative staff can access the updated appointment information.

Main Success Scenario (Basic Flow)

1. The administrative staff member selects the option to register a patient's arrival.
2. The system requests the option to search for a patient
3. The administrative staff member enters the patient's information.
4. The system identifies the patient and provides their scheduled appointments.
5. The administrative staff member selects the appointment for which the patient has arrived.
6. The system provides the appointment information and verifies that the appointment is eligible for arrival registration.
7. The administrative staff confirms the patient’s arrival
8. The system updates the appointment to show that the patient has arrived and saves the information
9. The system provides a message confirming that the patient's arrival was successfully registered.

Extensions (Alternative and Failure Flows)

5a. The patient does not have an appointment scheduled for the current day.

1. The administrative staff member cannot find an appointment scheduled for the patient on the current day.

2. The administrative staff member informs the patient that they do not have an appointment scheduled for today.
3. The use case ends without registering the patient's arrival.

Special Requirements

- Only authorized administrative staff can register patient arrivals and access the appointment information they are allowed to view.
- The system must record which administrative staff member registered the patient's arrival and when it was registered.
- The system must protect the patient's information from unauthorized access.

Technology and Data Variations List

- The administrative staff can search for the patient using their insurance number, date of birth, phone number, or partial first and/or last name.

Frequency of Occurrence

- Whenever the patient arrives for their scheduled appointment.

Miscellaneous

- **Open Issues:**
  - What happens if the patient arrives at the clinic but their appointment has already been cancelled by the administrative staff?
  - What happens if the patient’s arrival has already been registered in the system?
- **Assumptions:**
  - The appointment has not already been cancelled.
  - The patient's arrival has not already been registered in the system.
  - The system displays a confirmation message after successfully registering the patient's arrival.
  - The system displays all scheduled appointments belonging to the patient when the administrative staff searches for them.
  - The administrative staff can select the correct appointment from the patient's scheduled appointments.
  - The system updates the appointment information to show that the patient has arrived.

Use Case 3: Record Consultation

Scope: Appointment Tracking System

Primary Actor: Healthcare Professional

Stakeholders and Interests:

- **Healthcare Professional:** Wants to record the results of the consultation so they can access the information later and keep track of the patient's medical condition.
- **Patient:** Wants the results of their consultation to be recorded so the information can be used for future treatments.
- **Clinic:** Wants to keep the patient's clinical records organized and protected so authorized healthcare professionals can access them.

Preconditions:

- The patient must already be registered in the system.
- The patient must have an existing appointment with the healthcare professional.
- The patient's arrival must already be registered in the system.
- The healthcare professional must be able to access the patient's information to record the consultation.
- The healthcare professional must already be registered in the system, logged in, and authorized to record consultations.

Success Guarantee (Postconditions):

- The consultation record is successfully created and saved in the system.
- The consultation record is associated with the patient, their appointment, and the healthcare professional who conducted the consultation.
- The patient's clinical records are updated with the information recorded during the consultation.
- The consultation record contains the relevant medical information, which may include observations, diagnoses, treatment recommendations, and follow-up requirements.
- The consultation information is available to authorized healthcare professionals based on their roles.

Main Success Scenario (Basic Flow)

1. The healthcare professional requests the option to record a consultation.
2. The system provides the healthcare professional's scheduled appointments.
3. The healthcare professional selects the patient's appointment for which the consultation is being recorded.
4. The system provides the patient's information, appointment details, and the fields for recording the consultation.
5. The healthcare professional enters the consultation information, which may include observations, diagnoses, treatment recommendations, and follow-up requirements.
6. The healthcare professional submits the consultation information.
7. The system saves the consultation record and updates the patient's clinical records.
8. The system provides a message confirming that the consultation was successfully recorded.

Extensions (Alternative and Failure Flows)

7a. The system cannot save the consultation record:

1. The system encounters an error when saving the consultation information.
2. The system informs the healthcare professional that the consultation could not be saved.
3. The patient's clinical records are not updated with the failed consultation.
4. The use case ends without creating a consultation record.

Special Requirements

- Only authorized healthcare professionals can record consultations and access the patient's clinical information based on their roles.
- The system must record which healthcare professional created the consultation record and when it was recorded.
- The system must protect the patient's clinical information from unauthorized access.
- Authorized healthcare professionals must be able to access the patient's previously recorded consultation information.

Technology and Data Variations List

- No specific technology or data variations are defined in the project description.

Frequency of Occurrence

- Whenever a healthcare professional finishes an consultation with a patient.

Miscellaneous

- **Open Issues:**
  - What happens if the healthcare professional makes a mistake in the consultation record and can they modify the information after it has been saved?
  - What happens if the system fails to save the consultation record? Can the healthcare professional try again without losing the information?
  - Does the healthcare professional need to complete the consultation before recording the information, or can they record it during the consultation?
  - Is there any required information that the healthcare professional must enter before the consultation record can be saved?
- **Assumptions:** [Any assumptions not explicitly defined in the project description]
  - The system displays all scheduled appointments belonging to the healthcare professional.
  - The appointment is already connected to the correct patient and healthcare professional.
  - The healthcare professional records the consultation after conducting it.
  - The system displays the patient's information and appointment details when the healthcare professional selects the appointment.
  - The system provides fields for the healthcare professional to enter the consultation information.
  - The patient does not need to have previous consultation records in the system.

- The system saves the consultation information into the correct patient's clinical records.
- The system displays a confirmation message after the consultation has been successfully recorded.
- Prescriptions and follow-up appointments are handled through separate use cases when needed.
- If the system fails to save the consultation record, no consultation record is created and the patient's clinical records are not updated.